

Eastern Refining Limited Chittagong Refinery: what the loss distribution says

Eastern Refining Limited Chittagong Refinery; expected climate loss to 2050 \$118.9m (23.8% of \$500.0m value) under a delayed transition; 1-in-20 outcome \$316.9m; leading physical hazard: cyclone (HIGH).

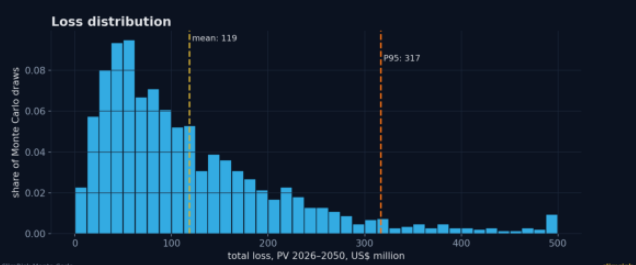
ClimRisk risk analyst · 2026-10-07

FINANCIAL RISK

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Financial risk · 07 October 2026

ClimRisk risk analyst; NGFS Phase 5 (IIASA)



JAS 2026. The Oceanic Niño Index reads +2.16 °C: El Niño. Since 1950 it has crossed +2 °C only 4 times [1]. The ocean is the loud part of the climate story this year. What follows is about the part that does not make the news: what climate policy and climate hazards do to the value of a specific set of industrial assets.

Eastern Refining Limited Chittagong Refinery in six numbers

The loss distribution, as modelled

\$500.0m

asset value

user-supplied value

\$118.9m

expected loss, delayed transition

Monte Carlo mean, PV 2026–2050

\$316.9m

1-in-20 outcome (P95)

Monte Carlo

\$407.6m

expected shortfall (worst 5%)

Monte Carlo

1.57%

physical annual loss, of value

IBTrACS × Emanuel; JRC / Deltares × FLOPROS

\$90.6m

physical share of the expected loss

present value 2026–2050

Source: ClimRisk risk analyst; NGFS Phase 5 (IIASA); JRC; Deltares; IBTrACS

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Monte Carlo means and percentiles; each source is stamped on the image.

0.3× quiet to loud

Here the loud risk is the larger: the quiet one is 0.3 times its size: \$29.3m of transition loss against \$90.6m of physical damage (delayed transition).

The loud risk: physical damage, priced

Physical damage is the risk that makes the news. Modelled average annual damage at Eastern Refining Limited Chittagong Refinery is **1.57% of value** (\$7.8m a year), and its present value to 2050 is \$90.6m. A 1-in-100-year event year costs \$219.1m, 44% of value. Leading hazard: cyclone (1.57% a year) [2][3].

Physical loss in a single year, by return period

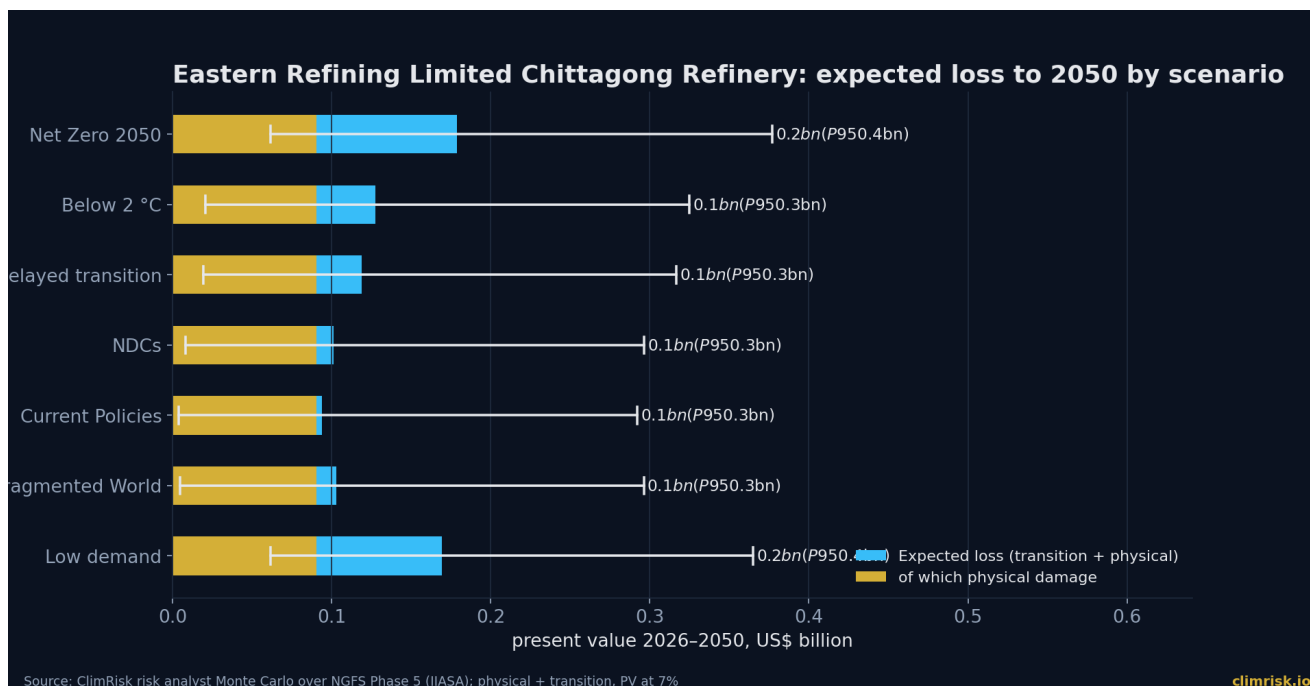


Source: ClimRisk risk analyst; IBTrACS × Emanuel (2011) × JRC flood × FLOPROS defences

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Single-year direct physical loss. Direct damage only [3][4].

The quiet risk: the transition, and what the mean hides

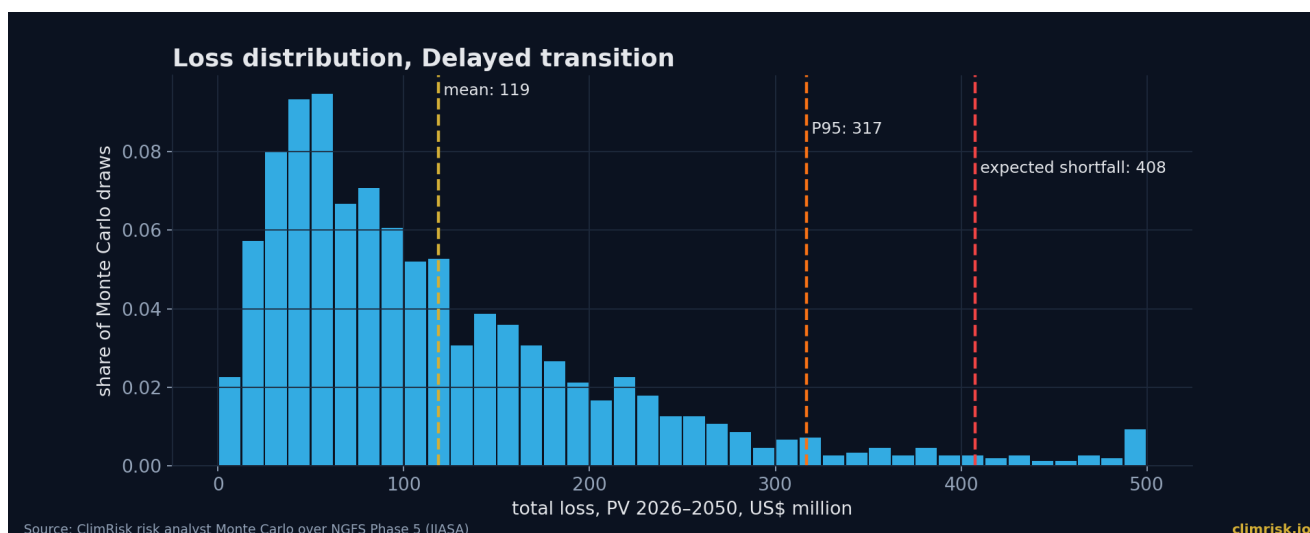


Gold: physical damage. Blue: total expected loss. Whisker: 5th to 95th percentile [5].

Physical: cyclone high — 0.44 tropical-storm-force events a year on average, 1-in-100-year wind 61 m/s (Cat 4). River flood negligible, coastal flood negligible; defences ≈ 1-in-50 (river) and 1-in-50 (coast). Modelled average annual damage 1.57% of value.

Policy: no explicit carbon price covers this asset today.

Transition (162,594 tCO₂e/yr): Net Zero 2050: \$69.6m carbon cost PV if unabated, \$51.3m with abatement, stranded \$38.8m; Delayed transition: \$20.4m carbon cost PV if unabated, \$17.1m with abatement, stranded \$12.1m; Current Policies: \$1.2m carbon cost PV if unabated, \$1.2m with abatement, stranded \$2.9m.



Share of Monte Carlo draws by total loss. The right tail is where lender covenants live.

Scenario	Expected loss	Share of value	Median	95th pct	99th pct	Expected shortfall
Net Zero 2050	\$178.9m	35.8%	\$156.6m	\$377.1m	\$500.0m	\$462.6m
Below 2°C	\$127.5m	25.5%	\$100.0m	\$324.9m	\$497.1m	\$416.6m
Delayed transition	\$118.9m	23.8%	\$90.7m	\$316.9m	\$481.0m	\$407.6m

Scenario	Expected loss	Share of value	Median	95th pct	99th pct	Expected shortfall
Nationally Determined Contributions (NDCs)	\$101.4m	20.3%	\$71.9m	\$296.4m	\$477.6m	\$392.0m
Current Policies	\$93.9m	18.8%	\$63.1m	\$292.3m	\$468.9m	\$386.1m
Fragmented World	\$102.8m	20.6%	\$74.8m	\$296.2m	\$485.5m	\$392.3m
Low demand	\$169.3m	33.9%	\$141.9m	\$364.9m	\$500.0m	\$451.0m

Present value 2026–2050 [5].

An industrial site is a set of structures whose failure modes depend on how it floods and how it is built, not on its sector label.

What inaction costs

Without any abatement the mean loss in the delayed transition would be \$122.1m, against \$118.9m with the sector levers: \$3.2m avoided.

1-in-100-year wind 61 m/s (Cat 4) — modelled average annual damage 1.57% of value. Verify roof, cladding and door ratings against the 1-in-100 wind; anchor rooftop and yard equipment; storm-mode shutdown procedure; review windstorm insurance limits and deductibles.

Carbon cost to 2050 (delayed transition, present value) \$20.4m if nothing is done (4% of asset value); with the sector's abatement levers deployed as prices rise \$17.1m (\$3.2m avoided). Electrify process heat, add CCS on hydrogen/SMR units, and plan throughput reduction in line with falling oil demand.

Modelled response: Efficiency, electrification, fuel-gas switching starts about 2035 (95% of runs by 2040); Capture on hydrogen units and catalytic cracking starts about 2044 (7% of runs by 2040). Get site-specific quotes for these levers (cost per tonne, lead time) and rerun with them in place of the library ranges; start permitting and procurement now for the long-lead options.

What this means, by reader

For the lender. The 95th percentile (\$316.9m) and expected shortfall (\$407.6m) are the covenant numbers, not the mean.

For the insurer. The physical share is a direct-damage figure; check deductibles against the single-year losses above.

For the investor. The distribution is wide; the decision depends on which scenario the plan assumes.

Method and limitations

1,500 Monte Carlo draws, discounted at 7%, over NGFS Phase 5 scenarios [5] for the transition and the engine's cyclone and flood models for the physical side [2][6][7]. Screening-grade model: validate asset values, defence standards and emissions with the owner before decisions. Direct physical damage only — business interruption and supply-chain losses are not included. Cyclone and river-flood frequencies are historical (no climate-change uplift); coastal depths include sea-level rise to 2050.

How ClimRisk measured this

- Physical → IBTrACS cyclone event set, JRC river and Deltares coastal depth, FLOPROS defences, damage curves.
- Transition → NGFS price and output paths, abatement levers, stranding.
- Monte Carlo → joint draws; expected loss, percentiles, expected shortfall.
- Value → the register's capacity-based estimate unless the operator supplies a value.

Sources

1. NOAA Climate Prediction Center, Oceanic Niño Index (ERSSTv5)
https://www.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ONI_v5.php
2. NOAA IBTrACS tropical-cyclone best tracks — 1980–2025 <https://www.ncei.noaa.gov/products/international-best-track-archive>
3. Emanuel (2011), Global warming effects on U.S. hurricane damage, *Weather, Climate & Society* 3:261 — wind damage sigmoid
<https://doi.org/10.1175/WCAS-D-11-00007.1>
4. Huizinga, de Moel & Szewczyk (2017), Global flood depth-damage functions, JRC — industrial curve
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5. NGFS Phase 5 climate scenarios (IIASA Scenario Explorer) — carbon price and sector output, 7 scenarios, 3 models
<https://data.ene.iiasa.ac.at/ngfs/>
6. JRC CEMS-GloFAS river flood hazard maps v2.1 <https://data.jrc.ec.europa.eu/collection/id-0054>
7. Deltares Global Flood Maps v2021.06 <https://deltares-research.github.io/global-flood-maps/>

Produced by the ClimRisk engine from the sources listed. Model outputs with the limitations stated; not forecasts or investment advice.